

CSEN 296B-2 | Week 8.1 Lab Submission

AstraNotes Collaborative Git Workflow

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Project: AstraNotes

1. Collaboration Log

This lab is completed as an individual simulation using two branches: main (stable baseline) and feature/search-improvements (focused change branch). The scenario is a search improvement: the existing search() method in NoteService had a silent bug where an empty keyword returned all notes instead of raising an error. The refactor also wrapped the return type in a SearchResult dataclass for traceability.

Step	Action	Branch	Notes
1	Confirm main is up to date	main	Baseline: 9 passing tests, note creation and sorted listing slices complete.
2	Create focused branch	feature/search-improvements	git checkout -b feature/search-improvements
3	Make change: refactor search()	feature/search-improvements	Empty keyword raises ValueError. Keyword stripped before match. Return type changed to SearchResult.
4	Add SearchResult dataclass	feature/search-improvements	Lightweight dataclass in note_service.py. Exposes .count and .keyword properties.
5	Write 8 new tests	feature/search-improvements	tests/test_search.py added. All 8 pass. 17/17 total tests green.
6	Commit with clear message	feature/search-improvements	See commit message below.
7	Open PR against main	PR: feature/search-improvements -> main	PR summary written. AI critique requested and reviewed.
8	Self-review of PR	Reviewer perspective	One issue found and fixed before merge (see review feedback section).
9	Merge PR	main	Merged after all 17 tests pass and review issue resolved.

2. Branch and PR Workflow Summary

Branch name: feature/search-improvements

Branch purpose: Refactor NoteService.search() to fix a silent failure mode (empty keyword returning all notes), add whitespace stripping, and wrap the return value in a SearchResult dataclass. Scoped to

one method and its tests only.

Commit message:

```
refactor(search): fix empty keyword bug and add SearchResult wrapper
```

- Empty or whitespace-only keyword now raises ValueError instead of silently returning all notes (FR-07 edge case, Week 3.1 gap review)
- Keyword stripped before comparison to avoid whitespace false-negatives
- Return type changed from List[Note] to SearchResult dataclass for traceability (keyword, notes, count accessible as named properties)
- 8 new tests added in tests/test_search.py, all passing
- No changes to other methods or other files

Traces to: FR-07, US-06

Tests: 17/17 passing

Files changed: src/astranotes/service/note_service.py (search method and SearchResult dataclass added), tests/test_search.py (new file, 8 tests).

Files not changed: all other modules, all other tests, requirements.txt, CLI entry point. The branch is tightly scoped.

3. Pull Request Summary

PR Title: refactor(search): fix empty keyword bug and add SearchResult wrapper

What this PR does:

Fixes a silent failure in NoteService.search() where passing an empty string returned all notes instead of raising an error. This violated the implicit contract of FR-07 (search shall return matching results for a keyword) and would have caused confusing behavior in the CLI if a user pressed Enter without typing a search term.

Also wraps the return type in a SearchResult dataclass so callers get structured output (keyword, notes list, count) instead of a bare list.

Why it is safe to merge:

- Change is limited to one method in one file
- 8 new tests cover the happy path, edge cases, and the fixed bug
- All 17 existing tests still pass (no regression)
- No new dependencies introduced
- SearchResult is a simple dataclass with no side effects

Why it is not yet complete:

- Private note search (decrypt-on-search path) is still deferred (documented as Gap 2 in Week 5.2 traceability matrix)
- Performance test for FR-07 (2-second target on 500 notes) not yet written

Traces to: FR-07, US-06

Reviewer: self-review (individual simulation)

4. Review Feedback

During self-review, one issue was identified before merge:

REVIEW COMMENT -- NoteService.search() return type change is a breaking change

File: src/astranotes/service/note_service.py, line search()

Issue: The return type changed from List[Note] to SearchResult. The CLI handler in cli/main.py calls search() and iterates the result directly as a list. After this change, iterating a SearchResult object would fail because SearchResult is not iterable.

Risk level: HIGH -- would cause a runtime AttributeError in the CLI before this is caught by any test.

Fix applied: updated cli/main.py search handler to use result.notes instead of iterating the return value directly.

```
# Before (broken after refactor):
results = service.search(keyword)
for note in results: # TypeError: SearchResult is not iterable

# After (fixed):
result = service.search(keyword)
for note in result.notes:
    print(f' - {note.title} (ID: {note.id})')
print(f' {result.count} result(s) for "{result.keyword}")')
```

This fix was committed as a follow-up before merge.

Status: RESOLVED

5. Merge Readiness Note

Check	Status	Notes
All tests pass	PASS	17/17 tests green including 8 new search tests
No regression in existing tests	PASS	9 original tests unchanged and passing
Breaking change in CLI fixed	PASS	cli/main.py updated to use result.notes before merge
Branch scoped to one idea	PASS	Only note_service.py, test_search.py, and cli/main.py changed
No new dependencies	PASS	requirements.txt unchanged
Traces to requirement	PASS	FR-07 and US-06 referenced in commit message and docstring
Private note search deferred	NOTED	Known gap documented in PR summary. Not a blocker for this merge.
Performance test for NFR-01	DEFERRED	2-second test on 500 notes not yet written. Deferred to Sprint 2.

Decision: Merge approved. All blocking checks pass. The two deferred items are pre-existing known gaps from the Week 5.2 traceability matrix, not new issues introduced by this branch. The branch is safe, scoped, and traceable.

6. Refactor Note

What changed and why:

Empty keyword behavior: The original search() returned all notes for an empty string. This was a silent failure mode that violated the contract of FR-07. The fix raises ValueError so callers are forced to handle the empty-input case explicitly rather than receiving unexpected results.

Return type: Changed from List[Note] to SearchResult. This is a small but meaningful improvement: the caller now knows what keyword produced the results and can display a count without re-computing len(). The tradeoff is that it is a breaking change for any caller iterating the return value directly, which is why the CLI fix was required.

Keyword stripping: A user typing ' sprint ' in the CLI should match the same notes as 'sprint'. The strip() call before comparison ensures whitespace in the input does not cause false-negative results. This addresses the same class of edge case as the whitespace-only title fix in ValidationLayer.

What was deliberately not changed:

The search algorithm itself (linear scan) was not changed. A performance improvement for NFR-01 is a separate concern and belongs in a separate branch.

Private note handling was not added. It is a more complex change requiring PrivacyService integration and is documented as Gap 2 in the traceability matrix.

No other NoteService methods were touched. The refactor was scoped to search() only.

7. How AI Helped

What AI helped with:

Suggested the SearchResult dataclass approach. The initial AI suggestion was to return a tuple (keyword, List[Note]). Changed to a dataclass because named properties are more readable and less error-prone than positional tuple access.

Generated the initial 8 test cases for test_search.py. The happy path and empty keyword tests were correct. Accepted with minor renaming for clarity.

Helped draft the PR summary structure: what the PR does, why it is safe, why it is not yet complete. The three-section format was AI-suggested and kept because it forces honest accounting of known limitations.

Identified the breaking change risk in the CLI when the return type changed. AI flagged this during a critique pass before the review section was written.

What was changed or rejected:

AI suggested adding a multi-keyword search feature (splitting on spaces and matching all terms). Rejected: this is a new feature, not a refactor. It belongs in a separate branch with its own requirement trace to FR-07.

AI suggested using a named exception SearchKeywordError instead of ValueError. Evaluated and rejected: ValueError is the correct Python convention for invalid argument values. A custom exception would add a class with no additional behavior.

AI suggested adding a test for search performance (timing assertion). Rejected for this branch: performance testing belongs with NFR-01 work, which is a separate and larger scope than this refactor.

AI did not suggest the keyword stripping fix. Added manually after reviewing the Week 3.1 ambiguity review, which identified whitespace handling as a recurring edge case pattern across the codebase.